

1. You may recall that we recently learned the derivative of $y = \sec(x)$.
 - a) Without referencing your notes or other sources attempt to prove/derive the derivative rule for $y = \sec(x)$.
 - b) What are some of the benefits of knowing how to derive a derivative rule versus just having it memorized? What are some of the benefits of having it memorized versus always having to derive it?
2. Show that if $f(x) = \frac{\sin(x)}{1 + \cos(x)}$ then $f'(x) = \frac{1}{1 + \cos(x)}$.

3. Determine the equation of a line which is perpendicular to $y = \frac{\tan(x)}{1 + \tan(x)}$ at $x = \frac{\pi}{4}$.
4. Suppose that $f(x) = \csc(x)$. For what values of x will the slopes of tangent lines on $y = f'(x)$ be horizontal on $[0, 2\pi]$.

5. Evaluate $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + h\right) - \sin\left(\frac{\pi}{3}\right)}{h}$.

6. Determine all values of x at which the slopes of tangent lines on $f(x) = \sin(x)$ and $g(x) = \cos(x)$ are opposite values.

7. Show that the function $f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is differentiable at $x = 0$. *Hint:* You need to use the definition of the derivative.